

UNIT 0 : Environment Setup — Virtualization & Linux Installation (3 Classes)

Lec 1 : Understanding Virtualization + Installing Oracle VirtualBox

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	What is Virtualization? Types of Hypervisors	10	40	Install Oracle VirtualBox on your personal laptop and download the Ubuntu LTS Desktop ISO file for the next class.	
2	VirtualBox vs VMware vs WSL — Overview	5			
3	Checking System Requirements (BIOS Virtualization, RAM, Disk)	5			
4	Downloading & Installing Oracle VirtualBox	15			
5	Exploring the VirtualBox Manager Interface	5			

Lec 2 : Creating a Virtual Machine & Installing Ubuntu

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	VM Configuration Concepts — RAM, vCPU, Disk Allocation	8	40	Complete the Ubuntu installation inside VirtualBox and take a screenshot of your desktop for submission.	
2	Creating a New VM & Attaching the Ubuntu ISO	10			
3	Live Demo: Ubuntu Installation Walkthrough	12			
4	Hands-on: Begin Ubuntu Installation	10			

Lec 3 : Post-Installation Setup, Guest Additions & First Terminal

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	Installing Guest Additions & Configuring Shared Folders	12	40	Take a snapshot named 'Fresh Install', open a terminal window, and run <code>whoami</code> and <code>pwd</code> — note the output.	
2	Snapshots — Create & Restore	10			
3	Opening Terminal + First Commands (<code>whoami</code> , <code>pwd</code> , <code>date</code>)	13			
4	Setup Recap Checklist	5			

UNIT 1 : Computer Fundamentals, OS & Introduction to Linux (3 Classes)

Lec 4 : Computer Fundamentals

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	What is a Computer? Hardware vs Software	10	40	Identify and list all hardware components inside your own computer/laptop (CPU, RAM, Storage type, etc.) with specifications.	
2	CPU, RAM, Storage & I/O Devices	15			
3	What Happens When You Open an App? (Disk→RAM→CPU→Output)	10			
4	Class Discussion & Real-World Examples	5			

Lec 5 : Operating System Fundamentals

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
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1	What is an OS? Why Do We Need One?	10	40	Prepare a one-page note explaining, in your own words, the major functions of an OS with real examples from your own device.	
2	Major Functions: Process / Memory / File / Device Mgmt	15			
3	Security & Protection + User Interfaces (CLI vs GUI)	10			
4	Recap Quiz	5			

Lec 6 : Linux Introduction & Architecture					
#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	What is Linux? Kernel vs Distribution	8	40	Explore the Linux terminal and identify: current user, current directory, OS information and kernel information. Submit terminal output as a text file.	
2	Linux vs Windows/macOS, Open Source, Where Linux is Used	10			
3	Linux Architecture: Kernel, Shell, User Space	10			
4	Hands-on: whoami, pwd, date, uname, clear, ls	12			

UNIT 2 : Linux Terminal, Files & File System (4 Classes)

Lec 7 : Terminal & Shell Basics + Navigation					
#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	Terminal vs Shell, Introduction to Bash	10	40	Navigate through /home, /home/student, Documents and Downloads using only the terminal and record every command you used.	
2	Command Structure: command [options] [arguments]	5			
3	Hands-on: pwd, ls, cd	15			
4	ls Options: -l, -a, -la	10			

Lec 8 : Creating & Managing Files / Directories					
#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	mkdir & touch — Creating Directories and Files	10	40	Create the folder structure College/Java, Linux, DSA, Projects using only terminal commands, then rename and move files between them.	
2	cp & mv — Copying, Moving & Renaming	10			
3	rm & rmdir — Deleting Files & Directories	10			
4	Hands-on: Build the College/ Folder Structure	10			

Lec 9 : Reading & Writing Files + Redirection					
#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	cat, less, head, tail, echo	15	40	Add content to notes.txt using echo and redirection inside each folder you created earlier.	
2	Redirection Operators: >, >>, <	10			
3	Hands-on: Writing & Appending to Files	15			

Lec 10 : Linux File System & Searching					
#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	Linux Directory Structure (/ , /home, /etc, /var, /usr, /dev, /proc)	10	40	Use find and grep to search for a specific file and word inside your College/ folder, and save the search results into output.txt.	
2	find — Searching for Files	10			
3	grep — Searching Text Inside Files	10			
4	Hands-on Lab: Search & Redirect Challenge	10			

UNIT 3 : Permissions, Users, Processes & System Management (4 Classes)

Lec 11 : Linux Users					
#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	User Concept — Root vs Normal User, User ID & Group ID	18	40	Run whoami, id, groups and who on your system and note down the output for each command.	
2	Commands: whoami, id, groups, who	12			
3	Hands-on Exercise: Identify Your User Details	10			

Lec 12 : File Permissions					
#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	Understanding ls -l Output & Permission Strings	12	40	Check permissions on 3 different files using ls -l and explain each permission set (owner, group, others).	
2	Read, Write, Execute (rwx) Explained	10			
3	Hands-on Lab: Permission Inspection Exercise	18			

Lec 13 : chmod & Ownership					
#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	chmod — Symbolic Mode & Numeric Mode (4-2-1)	15	40	Create a script file, make it non-executable, then use chmod (symbolic and numeric) to make it executable. Change owner/group with chown/chgrp using sudo.	
2	chown & chgrp — Changing Owner and Group	10			
3	Hands-on Lab: Permission Modification Challenge	15			

Lec 14 : Processes & System Information					
#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	Program vs Process, Commands: ps, top, htop	12	40	Start a background process (e.g. sleep 100 &), find its PID using ps/top, then terminate it using kill. Also check system RAM and disk usage.	
2	kill, jobs, fg, bg — Managing Processes	10			

3	System Info: uname, uptime, free, df, du	10			
4	Hands-on: Start, Find & Kill a Process	8			

UNIT 4 : Shell Scripting, Pipes, Networking & Package Management (6 Classes)

Lec 15 : Bash Scripting Basics

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	Writing Your First Bash Script (#!/bin/bash), chmod +x	12	40	Write your first Bash script that prints 'Hello Linux' and run it after making it executable.	
2	Variables & Command Substitution	13			
3	Environment Variables	8			
4	Hands-on: Hello Linux Script	7			

Lec 16 : User Input

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	Reading User Input with read	10	40	Write a Bash script that takes a student's name and marks as input and displays a formatted result on screen.	
2	Hands-on: Take Name & Marks as Input	20			
3	Discussion & Testing	10			

Lec 17 : Conditional Statements

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	if / else Syntax	12	40	Write a script to check whether a number entered by the user is even or odd.	
2	Comparison Operators (-eq, -ne, -gt, -lt, -ge, -le)	10			
3	Hands-on Lab: Even/Odd Checker Script	18			

Lec 18 : Loops & Functions

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	for Loop	10	40	Write a script to print numbers from 1 to 10 using a loop.	
2	while Loop	10			
3	Writing Functions in Bash	10			
4	Hands-on: Number Printer Script	10			

Lec 19 : Pipes & Redirection

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	Pipes — Connecting Command A to Command B	10	40	Use pipes to search for a specific running process by name (e.g. ps aux grep bash).	

2	Redirection Recap: >, >>, <,	8			
3	Hands-on: ps aux grep	12			
4	Challenge / Discussion	10			

Lec 20 : Networking & Package Management

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	Basic Networking Concepts (IP, Hostname, Client-Server)	8	40	Check your system's IP address, test connectivity with ping/curl, and install/remove a package using apt.	
2	Commands: ip addr, ping, curl, wget	12			
3	Package Management: apt update / install / remove & sudo	12			
4	Hands-on: Network Check + Package Install/Remove	8			

UNIT 5 : Linux Administration, System Calls, Automation & Mini Project (5 Classes)

Lec 21 : System Administration, Services & Logs

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	Root Privileges & sudo	8	40	Use systemctl status to check a running service, then use tail/grep on /var/log/syslog to find recent log entries.	
2	Services & systemctl	12			
3	Logs & /var/log	10			
4	Hands-on: Service Check + Log Inspection	10			

Lec 22 : System Calls & Process Creation

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	Application → System Call → Kernel → Hardware Flow	10	40	Explain in your own words the flow: Application → System Call → Kernel → Hardware, with one real example.	
2	Key System Calls: open, read, write, close	10			
3	fork(), exec(), wait() — Process Creation Concepts	12			
4	Parent Process vs Child Process	8			

Lec 23 : Automation & Scripting Practice

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	Why Automation Matters (Manual vs Scripted)	8	40	Write a shell script (system-info.sh) that automatically displays Username, Hostname, Kernel, RAM, Disk and Uptime when executed.	
2	Hands-on Lab: Build system-info.sh Script	25			
3	Recap & Doubt Clearing	7			

Lec 24 : Mini Project — Planning & Development

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	Project Options Overview (Student Mgmt / Sys Info / Backup Tool)	8	40	Choose one mini project (Student Management System / Linux System Info Tool / Backup Automation Tool) and begin building it.	
2	Planning Your Script — Variables, Functions, Loops Needed	12			
3	Hands-on Lab: Project Development Time	20			

Lec 25 : Mini Project — Testing & Presentation

#	Topic	Duration (min)	Total Lec Time	Assignment / Practical	
1	Testing & Debugging Your Script	15	40	Complete testing of your mini project and demonstrate it live in the terminal to the class.	
2	Live Project Demonstration by Students	20			
3	Course Wrap-up & Final Q&A	5			